A DURATION STUDY OF CHILDREN'S NEUTRAL TONE ACQUISITION IN MANDARIN

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ABSTRACT

The neutral tone in Mandarin is a very important part, and children's neutral tone acquisition should also be paid attention. This study focuses on the duration of the neutral tone. According to phonetic production recorded from children's daily dialogue, we examine the length of its duration through three aspects, namely the position of the neutral tone, the words of different types with the neutral tone, and the neutral syllable with different vowels, and have drawn that D (middle) < D (end), D (measure word) < D (modal particles) < D (bi-multi-syllable words), and D (X+e5) < D (X+u5) < D (X+ii5) < D (X+a5). Furthermore, we discuss possible factors which influence its duration and the reasons why such phenomena happen.

***Index Terms*—** neutral tone, duration, acquisition

1. INTRODUCTION

Mandarin has four lexical tones, high-level(55), rising(35), dipping(214) and falling(51). Tone is one of the most important part in the process of phonological acquisition in Mandarin. Besides, tone sandhi and neutral tone also should not be ignored in studies. Especially in children's acquisition, appearance of such phenomena is obvious along with that of vowels, consonances, and lexical tones.

1.1. Review in children's tone acquisition

Although studies of tone acquisition are conducted later than those of segments. In recent years, many studies have paid special attention to children tone acquisition in Mandarin. Most of those studies drew a similar conclusion that the acquisition of tones precedes that of segment, and contour tone acquisition is posterior.

Qi and Chen (2004) refer to a study conducted among Taiwan children by Li and Tang, whose results show that Mandarin-speaking children acquire firstly tones, and then vowels and consonants. Li, Zhu, Dodd etc. (2000) find that no errors in tones exist among children ranging from 1,6 to 4 years old. Zhu (2002) elaborates this conclusion by a table of age of stabilization of tones, which shows that not long until 1,9.15 children could acquire all lexical tones. However, Yang (2006) maintains that children could not acquire most of lexical tones before 2,3 years old.

Regarding tone sandhi, Zhu (2002) conducts certain statistical study of tone sandhi, and have the results that the process of tone sandhi acquisition, which includes morphologically conditioned tone sandhi, is approximately from 1,4 to 1,11. Wang, Schmitt, and Lin (2012) point out that a developmental pattern in children's acquisition of Tone 3 sandhi in flat structure, and suggest that foot-building in digits does not always follow the typical foot-building process.

1.2. Neutral tone and that of children's acquisition

Pervious studies have mainly concerned with issues like lexical tone, tone sandhi, and perception both acoustically and phonologically. However, few studies have examined children's neutral tone acquisition. The neutral tone acquisition might emerge synchronously along with other phonological acquisition phenomenon. It is of significance to study what the children's neutral tone is like.

Phonologically determined, neutral tone is a particular type of tone sandhi. Former studies are focused on Beijing dialect and neutral tones in bi-syllables. Lin and Yan (1980) maintains that duration is the main phonetic element of the neutral tone, and the duration is relatively 1/2 of that of lexical tones. Wang (2003) points out in the aspect of tones, the neutral tone only has one mora but no tone features. Feng, Chu, He etc. (2001) conduct a statistical study of tone duration, and get the results that the duration of neutral tone is relatively short than other tones (See Figure 1).

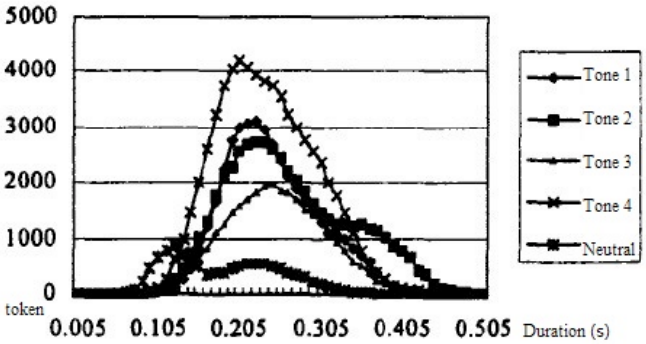

Figure 1: *Distribution of duration of different tones in Mandarin*

Regarding children's neutral tone acquisition, it is very possible that children's neutral tone has different appearance compared with that of adults, such as whether the position of the neutral tone could influence the duration is open to our study. Meanwhile, different types of neutral tone words also exist, as suffixed word ("Bei1Zi5", cup), double-barrelled words ("Ma1Ma5", mother) and others. It is possible that those have their particular differences in duration from others. Furthermore, as those different types of neutral tone words mentioned above, whether the syllables with different consonants and vowels would influence the duration of the neutral tone compared with the regular is open to study as well.

This paper focuses on the duration of neutral tone in children's acquisition. Three questions in our study are:

1) what is the children's neutral tone duration like?

2) as different types of the neutral tone words exist, what are differences of the duration between those types?

3) does the duration of the neutral tone have certain relationship with the syllables with different vowels?

2. METHODOLOGY

2.1. Participants

NAME	AGE
MYL	1,8
SYC	1,8
WM	1,9
LHR	2,0
WQC	2,2
LJX	2,3
FXX	2,4
LMH	2,4
WXX	2,6
KAS	2,10
FZR	3,0

Table 1: *Participants' Information*

12 monolingual Mandarin-speaking children (See Table 1) aged from 1,8 to 3 years old participated in this experiment. They are all growing up in Beijing, and acquiring Mandarin as their native language. No impairment of hearing or speaking is found among them as well. Besides, all of their parents speak Mandarin to them.

2.2. Recording and stimuli

12 participants accompanied by their parents were invited to a sound-proof chamber. In front of them, there is a screen showing a PPT which contains target stimuli. Equipped with professional recording apparatus, they were asked by their parents in dialogues to answer questions, such as what it was on the screen. In order to encourage children to correctly reply the PPT, possible questions raised by their parents might be "Zhei4ge5 Shi4 Shen2me5 Ya1?" (What's this?), or "Zhei4ge5 Shi4 Yong4lai2 Zuo4 Shen2me5 De5 Ya1?" (What's this used for?) and so on.

Target stimuli are some basically-used words, including numbers, colors, animals, goods, cartoons, and movements, and they are shown as the form of the picture or the cartoon to the children (See table 2).

Cup	Egg	Tian'anmen
Dog	To wear pants	Foot
Car	Ballon	Watermelon
Cap	Xi'yangyang	Grandfather
To fly a kite	To read a book	Socks
Basin	Banana	Dress
Hot	To brush teeth	Three (times)
Red/yellow/blue	The moon	Round
Star	On the roof	Wangzai milk
Father/mother	Child	Potato chips
Cabbage	Panda	

Table 2: *Pictures shown on the PPT*

2.3. Data processing

Sounds were recorded separately and saved as *.wav files. We use Praat to process those data, creating TextGrid with several tiers to annotate the sounds with the boundaries of syllables and phonemes.

Besides, the errors in both consonants, vowels and tones were particularly labeled. We pay special attention to study those errors in consonants or vowels with the neutral tone. Moreover, those overlapped and noised sounds were also labeled for not being used in the analysis. After that, we collect all of the available segments of sounds and extract the duration of each segment, and get 585 valid sample data of the neutral tone's duration.

3. DURATION ANALYSIS AND DISCUSSION

3.1. The duration of the neutral tone at different positions

With those 585 sample data of the neutral tone's duration, we analysis the duration of the neutral tone in children's acquisition (See Table 3).

	Mean (ms)	Standard Variation	Sample number
The duration of the neutral tone	180.19	92.73	585

Table 3: *The duration of the neutral tone in children's acquisition.*

Wang (2003) did an experiment to study the prosody of Mandarin spontaneous speech, in which used were materials that corpus of Chinese adults' telephone recordings (Zong, 1999). He found that the position of the neutral tone might influence the duration, and the neutral tone syllables at the end are relevantly long. Wang's results are in Table 4 above.

If we put these two results together, we could make a compare between children's neutral tone acquisition and adults'. However, we find that the position of the neutral tone is not included as one consideration in our study, which may explain why the standard variation of our data seems too high compared with Wang's results.

	Mean (ms)	Standard Variation	Sample number
The duration of the neutral tone syllable in the middle	109.3	45.3	439
The duration of the neutral tone syllable at the end	185.4	62.4	398

Table 4: *The duration of the neutral tone syllables (Wang, 2003).*

We suppose that the position of the neutral tone in children's acquisition is similar to that of adults. Thus, in order to further study the duration of the neutral tone, and make a peer-to-peer compare, we classify all the neutral tone words according to their positions (See Figure 2). We get new results as below (See Table 5 and Figure 3).

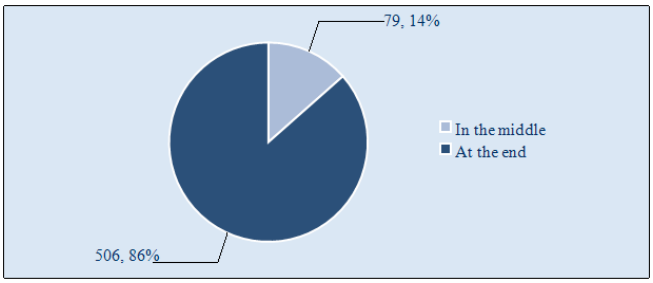

Figure 2: *Percentage of the neutral tone words with different positions.*

As we could see in these table and figures, the results seem not as what we thought. Firstly, the difference in sample number is huge with only 13.5% of the neutral tone syllables are in the middle of a sentence. That is because in the data we collect, children are asked to reply to certain pictures with some simple sentences or just several words. Secondly, we could simply reach that compared the duration mean with Wang's results, children's duration of the neutral tone syllable at the end is close to that of adults, and D(at the end) > D(in the middle), whereas the durations of the neutral tone syllable in the middle in both two results vary about 50 ms. Thirdly, we find that the neutral tone syllables in the middle are shorter (86%) than those at the end. Wang (2003) points out that it's nearly 1.7:1 between the neutral tone syllables in the middle and at the end. Therefore, we see that for children, the gap is not as large as that of adults. Fourthly, the standard variation of that at the end does not decline as we suppose to be. Such a high standard variation reveals that children's neutral tone acquisition has not be as stable as that of adults, and those neutral tone syllables in the middle might be stabler than those at the end.

	Mean (ms)	Standard Variation	Sample number
The duration of the neutral tone syllable in the middle	158.09	78.25	79
The duration of the neutral tone syllable at the end	183.64	94.40	506

Table 5: *The duration of the neutral tone in children's acquisition, according to the position of the neutral tone words.*

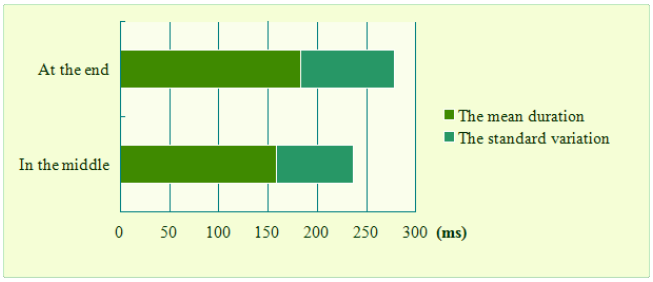

Figure 3: *The mean durations and the standard variations of the neutral tone words with different positions.*

Although the influence of the different positions in the neutral tone words is obvious in the adult group (Wang, 2003), however, it's hard to claim that similar influence happens in children's neutral tone acquisition with a quite high standard variation. The position of the neutral tone is not the only one important factors influencing the duration in children's acquisition.

3.2. The duration of the neutral tone according to different types of the words

As explained above, the supposition that the position of the neutral tone syllable is not applicable to the children's neutral tone acquisition. Therefore, we look back to our data, and find that the neutral tone words themselves might be classified as different types as their patterns vary from each other. According to our recording and data, we classify the neutral tone words into five different types, the first the suffixed word ("Bei1Zi5", cup; "Pa1Zhe5", grovel), the second the reduplicated words ("Ma1Ma5", mother; "Lao3Ye2Ye5", grandfather; "Gou3Gou5", dog), the third the words with measure words ("Wu3Ge5", five), and the fourth some certain bi/multi-syllable words ("Yi1Fu5", clothes; "Feng1Zheng5", kite); the fifth the modal particles ("Chi1Fan4Ma5", interrogative mood; "Lian4Gong1Ne5", exclamatory mood). Thus, from a new perspective, we

suppose that different types of the neutral tone words might also have certain differences upon the durations. To confirm our supposition, we get the classified data as shown below in Table 6.

	Mean (ms)	Standard Variation	Sample number
The duration of the neutral tone syllable of the suffixed words	170.96	100.50	209
The duration of the neutral tone syllable of the reduplicated words	219.08	111.97	75
The duration of the neutral tone syllable of the words with measure words	147.80	60.94	52
The duration of the neutral tone syllable of some bi/multi-syllable words	190.92	78.74	109
The duration of the neutral tone syllable of the modal particles	173.15	77.68	139

Table 6: *The duration of the neutral tone in children's acquisition, according to different types of the neutral tone words.*

For the results in Table 6, we find that the percentage of each type of the neutral tone words vary very much (See Figure 4), and thanks to the different types, we could now further tell the differences of the neutral tone words (See Figure 5).

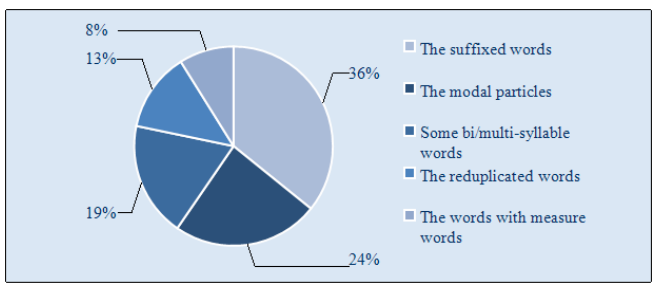

Figure 4: *Percentage of different types of the neutral tone words.*

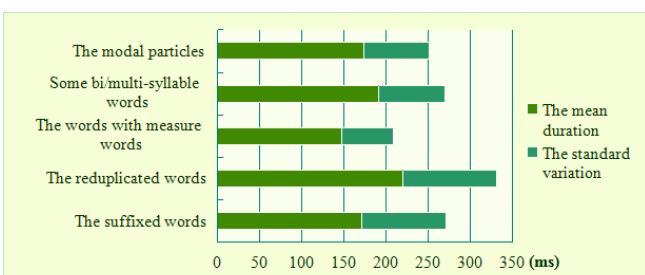

Figure 5: *The mean durations and the standard variations of different types of the neutral tone words.*

As shown above, we find each standard variation vary, especially the suffixed words and the reduplicated words with large variations. There are two main reasons to explain it. Firstly, such classification is not precise enough, as "X+Zhe5", "X+locatives", and "X+Zi5" are all included in the category of the suffixed words; secondly, the durations of the neutral tone words vary because of different syllables with all possible combinations of consonants and vowels. Therefore, the neutral tone durations of the suffixed words and the reduplicated words are not valid and convincing with the standard variations above 100.

For other three types, we could find that the shortest mean duration of the neutral tone is of the words with measure words, such as "X+Ge5", with also a smallest standard variation; the second is of the modal particles, usually at the end of a sentence with an intonation with certain types; the last is of some bi/multi-syllable words, such as "Dong1Xi5" and "Xi3Huan5". It's quite interesting to find such an order. From a perspective of principle of information quantity, from a measure word to a modal particle, and then to a bi/multi-syllable word, information quantities increase. Thus, it seems that a given word with higher information quantities tends to be elongated in their durations. Also, the standard variation of the neutral tone of bi/multi-syllable words reveals that the durations of those words are not that stable. Considering the type, it is possible that in children's acquisition, neutralization is a process, which is from the original tone to the neutral, and for those words, children have not yet pronounced the target neutral tone words stably. What's more, for the modal particles, intonation with different types added, such as angry, happy, and sad, might possibly bring other factors into the neutral tone. Thus, its standard variation is also relevantly high.

3.3. The duration of the neutral tone according to the words with different vowels

In 3.2., we fail to get valid results about the neutral tone words in form of the suffixed and the reduplicated due to rough classification. As we know lexicological classification cannot distinguish phonetical differences, and we consider, those syllables with different vowels might influence their durations. In order to further discuss, we suppose that within the suffixed and the reduplicated neutral tone words and even over all the neutral tone words, syllables with different vowels might have different durations. In our data, we find some neutral syllables appear much frequently, as "e5", "ii5", "a5", and "u5" (all are the regular vowels).

Therefore, we count and percentile those syllables with certain vowels as in Figure 6 below, and get the duration results as shown in Table 7 and Figure 7.

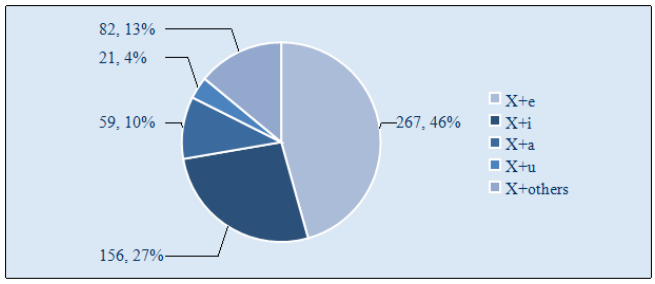

Figure 6: *Percentage of different syllables with certain vowels of the neutral tone words.*

	Mean (ms)	Standard Variation	Sample number
X + e5	151.25	63.51	267
X + ii5	184.73	89.41	156
X + a5	188.78	61.34	59
X + u5	167.20	76.96	21
X + others	189.79	76.82	82

Table 7: *The duration of the neutral tone in children's acquisition, according to different syllables with certain vowels.*

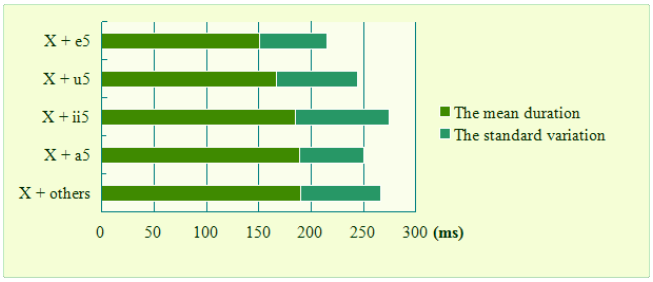

Figure 7: *The mean durations and the standard variations of different syllables with certain vowels.*

As shown in Table 6 and Figure 7, we can clearly find the duration order for the neutral tone, D (X+e5) < D (X+u5) < D (X+ii5) < D (X+a5), which shows that syllables with different vowels may influence the durations of the neutral tone. Moreover, for their standard variations, we find that X+ii5 has a high variation, 89.41, compared with syllables with other vowels. The reason why the phenomenon happens is that there are several errors of the vowels, according to our data, such as "X+i5", "X+iii5", "X+e5" instead of "X+ii5", whereas other vowels do not have as many errors as of it. Thus, we consider, in children's acquisition of the neutral tone, the duration of the syllable "X+ii5" is not as stable as others due to the influence of the vowels.

4. CONCLUSION

The study adopts statistical analysis of the duration of the neutral tone in Mandarin children's acquisition with three points. 1) There are many factors that might influence the duration of the neutral tone, such as its position, type, and vowels. 2) For the positions of the neutral tone, the duration of that in the middle seems shorter than that at the end, D (in the middle) < D (at the end). However, since the standard variations is quite high, we hardly could draw that in children's acquisition, the duration in the middle is shorter than that at the end. 3) For the types of the words with the neutral tone, different types of the words have different durations. The duration order between three types of the words with the neutral tone is that D (measure word) < D (modal particles) < D (bi-multi-syllable words). However, the types, which are the suffixed and the reduplicated words with the neutral tone is not enough to study the duration with the high standard variations above 100. 4) For the syllables with different vowels, we also get an order according to different vowels, D (X+e5) < D (X+u5) < D (X+ii5) < D (X+a5).

Some questions remain open to study after this simple trial of statistical duration study. 1) How vowels and consonants influence the duration of the neutral tone? 2) Whether the errors would influence it? 3) Does children have certain awareness of the neutral tone? 4) How children percept the neutral tone according to their duration?

5. REFERENCES

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